

LAB INCHARGE / SYSTEM ADMINISTRATOR

Lab Setup & Prerequisites

Two tiers: what is essential, and what is a bonus

Workshop: Agentic AI for Teaching & Academic Processes

Institute: Symbiosis Institute of Technology, Pune

Facilitator: Mr. Dinesh Wadhwani

Workshop dates: 10–11 September 2026

Tier 1 deadline: Monday 7 September

Tier 2 deadline: Tuesday 8 September

Machines: 40 (approx.)

Read this first

This document is for the **Lab Incharge / System Administrator**. It covers everything needed on the lab machines before the Agentic AI workshop on **10-11 September 2026**.

The good news

The core workshop needs almost nothing installed. Everything essential runs in a browser: Apify's cloud IDE, Google AI Studio, and Gemini. If you install nothing at all beyond a working browser, the workshop still runs at full value.

The installation work in this document falls into two tiers. Please treat them differently.

Tier	What	If it fails
TIER 1 — Essential	Browser, network access, VS Code	Workshop cannot run. Escalate immediately.
TIER 2 — Optional	Node.js, n8n, Apify CLI	Some comparison exercises are dropped. Workshop runs fine without them.

PLEASE DO TIER 1 COMPLETELY BEFORE STARTING TIER 2

Tier 1 takes about fifteen minutes per machine and is what the workshop depends on. Tier 2 is a bonus that enables a set of side-by-side comparison exercises.

If you run short of time, a lab with Tier 1 done properly on all forty machines is far better than a lab with both tiers done on twenty-five.

Timeline

Day	Task
Monday 7 Sept	Tier 1 on all machines. Network verification. Report blockers same day.
Tuesday 8 Sept	Tier 2 if Tier 1 is complete and verified.
Tuesday evening	Confirm handover checklist to facilitator.

DO NOT INSTALL ON THE MORNING OF THE WORKSHOP

Forty machines downloading packages simultaneously over campus Wi-Fi will saturate the link.

Work in staggered batches of eight to ten machines.

IF YOUR LAB USES DISK IMAGING

Build one machine completely, verify it against Section 6, then clone. This turns eight hours of work into ninety minutes and guarantees all forty machines behave identically — which matters enormously during a live session.

DEEP FREEZE OR RESTORE-ON-REBOOT SOFTWARE

It must be **thawed** before installing and must **remain thawed** for 10–11 September, or installations vanish on reboot and participant work is lost between sessions.

Please confirm this with the department before starting.

TIER 1 — Essential

1.1 Machine requirements

Requirement	Minimum
Operating system	Windows 10 64-bit or later
RAM	8 GB
Free disk space	5 GB (Tier 1) / 10 GB (with Tier 2)
Browser	Google Chrome or Microsoft Edge, current version
Administrator rights	For installation only
Screen resolution	1366×768 minimum

1.2 Browser

Chrome or Edge, updated to current. Participants will run a cloud IDE in the browser, so an old build will cause odd editor behaviour.

Participants must be able to sign in to a personal Google account. If the browser is locked to institutional sign-in only, please raise this immediately — see the warning in Section 7.1.

1.3 Visual Studio Code

Used for editing code locally and viewing exported files.

1. Go to **<https://code.visualstudio.com>**
2. Download the Windows **System Installer**, 64-bit — not the User Installer, which installs per-profile
3. Run as Administrator
4. On "Select Additional Tasks", tick: - Add "Open with Code" to the file context menu - Add "Open with Code" to the directory context menu - **Add to PATH**
5. Install

Extensions (Extensions panel, `Ctrl+Shift+X`):

Extension	Publisher
Python	Microsoft
ESLint	Microsoft

Verify: `code --version` in Command Prompt returns a version number.

1.4 Network — the critical section

Please have the network team confirm the proxy and firewall permit **all** of the following.

Domain	Purpose	Tier
<code>apify.com</code>	Registration, Console	1
<code>console.apify.com</code>	Cloud IDE where all Actors are built	1
<code>api.apify.com</code>	Actor API calls	1
<code>aistudio.google.com</code>	API key generation	1
<code>generativelanguage.googleapis.com</code>	All AI calls	1
<code>gemini.google.com</code>	Gemini web app, Gems	1
<code>notebooklm.google.com</code>	Gemini Notebook	1
<code>code.visualstudio.com</code>	VS Code and extensions	1
<code>nodejs.org</code>	Node.js download	2
<code>registry.npmjs.org</code>	npm packages	2
<code>localhost:5678</code>	n8n editor	2

THE ONE THAT GETS MISSED EVERY SINGLE TIME

`generativelanguage.googleapis.com` is the endpoint that carries the actual AI requests.

It is very common for a proxy to allow `aistudio.google.com` — so the website loads perfectly and everything looks configured — while silently blocking the API endpoint.

The result: everything appears fine until the first AI call runs, which then fails with an obscure network error, mid-session, in front of the room, with no obvious cause.

Do not assume this works because the AI Studio page loads. Test it explicitly using the script in Section 6.2.

SECOND MOST MISSED

`console.apify.com` runs a full cloud code editor in the browser. Some content filters classify cloud IDEs as developer tools and block them.

Test it by actually opening the site and loading the editor, not just pinging the domain.

TIER 2 — Optional

Everything below enables a set of comparison exercises where faculty build the same agent on two platforms. Valuable, not essential.

2.1 Node.js

VERSION CONSTRAINT

n8n requires Node.js **between 20.19 and 24.x inclusive**. Anything outside that range will cause n8n to refuse to start or behave unpredictably.

Install Node.js 22 LTS.

1. Go to **<https://nodejs.org>**
2. Download the **LTS** Windows Installer (.msi), 64-bit
3. Run as Administrator
4. On "Tools for Native Modules", **leave the checkbox unticked** — it triggers a large Chocolatey and Build Tools download that is not needed and can take 30+ minutes
5. Install

Verify:

```
node --version  
npm --version
```

Expected: `v22.x.x` and `10.x.x`.

2.2 n8n

Pin one version across all machines

n8n releases a new minor version most weeks. Different versions in one room means workflow exports may not import cleanly between machines — very hard to diagnose live.

On the **first machine only**:

```
npm view n8n version
```

Write the number down. Use that exact version on every other machine.

Install

```
npm install -g n8n@2.9.2
```

Replace `2.9.2` with whatever the command above reported. Takes 8–15 minutes per machine and produces a lot of console output. Warnings are normal; only `npm ERR!` indicates real failure.

WHY NPM, DESPITE THE DEPRECATION NOTICE

n8n's documentation marks the npm installation method as deprecated from n8n 3.0 and recommends Docker Compose or its one-line setup.

Neither is workable here. The one-line setup requires Linux or macOS with curl. Docker on Windows requires WSL2, hardware virtualisation enabled in BIOS, and a multi-gigabyte download per machine.

The npm method works correctly on the current 2.x line and is appropriate for a two-day teaching workshop. This is a deliberate, informed choice — and it is Tier 2, so if it fails, nothing essential is lost.

First launch

```
n8n start
```

Wait for `Editor is now accessible via: http://localhost:5678`, then open that address in a browser.

n8n presents an **owner account setup screen** on first run. Use shared lab credentials:

Field	Value
Email	<code>lab@sitpune.local</code>
First / Last name	<code>SIT / Lab</code>
Password	One standard password for all machines — give it to the facilitator

Stop n8n with **Ctrl+C** once verified.

Desktop shortcut

Create `Start n8n.bat` on the Desktop:


```
@echo off
title n8n - do not close this window
echo Starting n8n. Please wait 30-60 seconds.
echo When ready, open http://localhost:5678 in your browser.
echo.
echo DO NOT CLOSE THIS WINDOW while using n8n.
echo.
n8n start
pause
```

CHECK THE FILE EXTENSION

If Windows is hiding extensions, Notepad saves this as `Start n8n.bat.txt`, which does nothing when double-clicked.

In File Explorer enable **View** → **File name extensions** and verify on every machine.

Firewall and port

Pre-approve the port, as Administrator:

```
netsh advfirewall firewall add rule name="n8n" dir=in action=allow protocol=TCP
localport=5678
```

Check the port is free:

```
netstat -ano | findstr :5678
```

No output means free.

2.3 Apify CLI

Optional. Lets faculty run Actors locally without a cloud build, which is faster for iteration. Requires Node.js from 2.1.

```
npm install -g apify-cli
```

Verify: `apify --version`

2.4 Python

Optional. Only needed if faculty choose Python Actor templates.

1. <https://www.python.org/downloads/> — version **3.11** or **3.12**
2. Run as Administrator

3. **Tick "Add python.exe to PATH" on the first screen.** Easy to miss, causes most Python problems in labs.

4. Install Now

Verify: `python --version` and `pip --version`

3. Known issue to be aware of

THE N8N PYTHON CODE NODE WILL NOT WORK — THIS IS EXPECTED

On Windows machines where n8n was installed via npm, the Python Code node fails with an error about a missing virtual environment. This is a known limitation of this installation method.

No action is required. Every n8n exercise in this workshop uses the **JavaScript Code node**.

This is documented here only so it is not mistaken for a broken installation and does not trigger an unnecessary reinstall.

4. Shared workshop folder

Please create a folder accessible from every lab machine — a network share or a folder copied locally to each Desktop.

Name it **Agentic-AI-Workshop** and place it on the Desktop or map it as a drive.

The facilitator will supply its contents by Tuesday evening. It will contain starter code, sample data files and the reference PDFs. Participants will be told to open it in Lab Test 0.

Please confirm the exact path to the facilitator so it can be printed on the board.

5. Time estimate

Task	Per machine
Tier 1	
Browser check	2 min
VS Code + extensions	6 min
Network verification	4 min
Tier 1 subtotal	~12 min
Tier 2	
Node.js	5 min
n8n install	8–15 min
n8n first launch + shortcut	6 min
Apify CLI	2 min
Python	4 min
Tier 2 subtotal	~25–35 min

Across forty machines in staggered batches: roughly **2 hours for Tier 1**, **6–8 hours for Tier 2**. With disk imaging, far less.

6. Verification

Do not mark a machine ready until the relevant checks pass.

6.1 Tier 1 check script

Save as `check-tier1.bat` on the Desktop and run:

```
@echo off
echo =====
echo   SIT Agentic AI Workshop - TIER 1 CHECK
echo =====
echo.
echo [1] VS Code
code --version
echo.
echo [2] curl available (needed for next script)
curl --version | findstr /i "curl"
echo.
echo =====
echo   Now run check-network.bat
echo =====
pause
```

6.2 Network check — the critical one

Save as `check-network.bat` and run:

```
@echo off
echo Testing required endpoints...
echo.
echo --- TIER 1 (must all work) ---
curl -s -o nul -w "generativelanguage.googleapis.com : % HTTP_CODE\n" https://
generativelanguage.googleapis.com
curl -s -o nul -w "aistudio.google.com : % HTTP_CODE\n" https://
aistudio.google.com
curl -s -o nul -w "console.apify.com : % HTTP_CODE\n" https://
console.apify.com
curl -s -o nul -w "api.apify.com : % HTTP_CODE\n" https://api.apify.com
curl -s -o nul -w "gemini.google.com : % HTTP_CODE\n" https://
gemini.google.com
echo.
echo --- TIER 2 (optional) ---
curl -s -o nul -w "registry.npmjs.org : % HTTP_CODE\n" https://
registry.npmjs.org
echo.
echo A three-digit number means the machine reached the server.
echo 000 or blank means BLOCKED - report immediately.
pause
```

6.3 Manual browser check

The script above confirms the domains resolve. It does not confirm the cloud IDE renders. On at least **five machines spread across the lab**, please also:

1. Open `https://console.apify.com` and confirm the page loads past the sign-in screen
2. Open `https://aistudio.google.com` and confirm it loads
3. Confirm you can sign in with a **personal Gmail** — not only an institutional account

6.4 Tier 2 check script

```
@echo off
echo [1] Node.js (need v20.19 to v24.x)
node --version
echo.
echo [2] n8n version (must match all other machines)
n8n --version
echo.
echo [3] Apify CLI
apify --version
echo.
echo [4] Python (3.11 or 3.12)
python --version
echo.
echo [5] Port 5678 free? (no output = free)
netstat -ano | findstr :5678
pause
```

PER-MACHINE SIGN-OFF

Tier 1 — all must pass:

- [] Chrome or Edge current
- [] Personal Google sign-in possible in the browser
- [] `code --version` returns a version
- [] `check-network.bat` returns three-digit codes for **all Tier 1 endpoints**
- [] `console.apify.com` loads the editor in the browser
- [] Shared workshop folder accessible

Tier 2 — nice to have:

- [] `node --version` shows v20.19 to v24.x
- [] `n8n --version` shows the **same** version as every other machine
- [] `Start n8n.bat` opens a working editor at localhost:5678
- [] `apify --version` returns a version
- [] `python --version` shows 3.11 or 3.12

7. Participant accounts — please circulate in advance

Created by participants themselves, not the lab team. **Circulating this in advance saves 25 minutes of the first session.** Please forward to all registered participants.

7.1 Google account and API key

INSTITUTIONAL GOOGLE ACCOUNTS WILL NOT WORK — PLEASE PASS THIS ON VERBATIM

Participants must sign in with a **personal Gmail address**, not their SIT credentials.

Google requires a Workspace administrator to enable a Gemini licence before certain features appear for work or school accounts. Faculty signing in with institutional credentials may find features **simply missing from the interface, with no error message** to explain why.

Anyone without a personal Gmail should create one before Wednesday. It takes two minutes and saves an hour.

Steps for participants:

1. Go to **<https://aistudio.google.com>** and sign in with a personal Gmail
2. Click **Get API key → Create API key**
3. Copy the key (begins `AIzaSy...`) and save it somewhere private
4. Send a test message in AI Studio to confirm the account works

The free tier requires **no credit card and no Google Cloud project**.

7.2 Apify account

Register here: <https://apify.com/?fpr=njgpfm>

1. Click **Sign up**, register with a **personal email address**
2. **No credit card required**
3. Verify the email and sign in to the Apify Console
4. Collect the API token: **Settings → Integrations → Personal API tokens**

The free plan includes **\$5 of platform credit each month**, 25 concurrent runs, and 7-day data retention. Ample for the workshop.

7.3 Security instruction for participants

PLEASE INCLUDE THIS TEXT WHEN YOU CIRCULATE

Both API keys grant access to your accounts. On shared lab machines:

- Do not email them, paste them into shared documents, or save them to the machine's Desktop
- Store the Apify token as an Apify **secret environment variable**, never typed directly into code that gets saved
- **At the end of Day 3, revoke both keys.** Gemini: aistudio.google.com. Apify: Settings → Integrations. Generating fresh ones for your own machine takes ten seconds.

Treat any key typed on a shared machine as compromised the moment you walk away.

8. Troubleshooting

Symptom	Cause	Fix
<code>console.apify.com</code> will not load the editor	Content filter blocking cloud IDEs	Escalate to network team — Tier 1 blocker
AI calls fail although AI Studio loads	Proxy blocking <code>generativelanguage.googleapis.com</code>	Escalate immediately — Tier 1 blocker
Cannot sign in with personal Gmail	Browser locked to institutional SSO	Escalate — Tier 1 blocker
<code>'code' is not recognized</code>	PATH not refreshed	Close all terminals, reopen. Reboot if needed.
<code>'node'</code> or <code>'n8n'</code> not recognized	PATH missing global npm path	<code>npm config get prefix</code> , add that folder to system PATH, reboot
n8n install fails, permission error	Not running as Administrator	Reopen Command Prompt as Administrator
n8n starts, nothing at localhost: 5678	Firewall blocked Node.js	Apply the netsh rule in 2.2
n8n port in use	Another service on 5678	Report the process to the facilitator
<code>Start n8n.bat</code> does nothing	Saved as <code>.bat.txt</code>	Enable extensions in Explorer, rename
n8n Python Code node errors on venv	Known limitation	Expected. All exercises use JavaScript. No action.
Deprecation warnings when n8n starts	Normal for current release	Ignore

9. Escalation

Issue	Escalate to	When
Any Tier 1 domain blocked	Network team	Immediately — no same-day workaround
Cloud IDE blocked by content filter	Network team	Immediately
Personal Google sign-in blocked in browser	Network team	Immediately
Machines lack admin rights	Department IT	Monday
Restore-on-reboot software active	Department IT	Monday
Fewer than 40 usable machines	Facilitator	As soon as known
Tier 2 not achievable in time	Facilitator	Tuesday — this is acceptable, just tell us

HANDOVER — PLEASE CONFIRM TO THE FACILITATOR BY TUESDAY EVENING

- ☐ Machines with **Tier 1** complete and verified: _____
- ☐ Machines with **Tier 2** complete and verified: _____
- ☐ All Tier 1 domains confirmed reachable
- ☐ generativelanguage.googleapis.com explicitly tested
- ☐ console.apify.com loads the editor on at least 5 machines
- ☐ Personal Gmail sign-in confirmed possible
- ☐ n8n version installed (if Tier 2): __
- ☐ Node.js version (if Tier 2): __
- ☐ Shared folder path: _____
- ☐ Restore-on-reboot confirmed disabled for 10–11 September
- ☐ Participant account instructions circulated
- ☐ Spare machines available: _____
- ☐ Lab contact reachable during sessions — name and number: __

A NOTE OF THANKS

The facilitator is aware this is short notice. Tier 1 is what matters; if Tier 2 does not happen, please say so on Tuesday rather than working late — the workshop is designed to run without it.